Lecture 9: Hypothesis Testing

Lecturer

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CLASS CODE

TOAE203

COURSE

Statistics and Probability in Economics and Business

Outline

01

INTRODUCTION

What a hypothesis test is for, and the five concepts the rest of the session depends on

02

TESTS FOR μ WHEN n IS LARGE

The full machinery: hypotheses, rejection region, the two errors, the p-value, and the five steps

03

TESTS FOR μ WHEN n IS SMALL

When the central limit theorem no longer applies, and the t distribution takes over

04

TESTS FOR A POPULATION PROPORTION

The same procedure applied to a percentage rather than a mean

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INTRODUCTION

Three Claims a Test Can Examine

Hypothesis testing uses sample data to examine a claim about a population

01

A production standard

Does an average box of cereal contain 368 grams?

02

A behavioral claim

Do FTU students spend more than 3 hours a day on the internet?

03

A signal in noise

A radio telescope records a varying amplitude. Is there a source, or is the variation only random noise?

INTRODUCTION

Key Statistical Concepts

Testing a hypothesis determines whether a claim about a population parameter is strongly supported by the sample

01

Null and alternative hypothesis

H_0 (giả thuyết không) and H_1 (giả thuyết thay thế): two competing statements, one of which will be chosen.

02

Type I and Type II errors

The two ways a decision can be wrong. They are not symmetric, and they cannot both be made small at once.

03

Level of significance

α: the probability of wrongly rejecting H_0, fixed by the analyst before the data are examined.

04

Rejection region

The set of values of the test statistic for which H_0 is rejected, bounded by a critical value.

05

P-value

How extreme the observed result is, computed on the assumption that H_0 is true.

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The ATM Problem

Transaction times at the bank have mean 270 seconds and standard deviation 24 seconds. An ATM is installed to reduce them. A sample of 38 transactions is taken.

How can the sample mean be used to test the claim that the true population mean transaction time is now less than 270 seconds? The number 270 is known, the new mean is not, and the only evidence available is 38 observations. Every element of the machinery built in this part exists to turn that evidence into a decision.

Formulation of H_0 and H_1

When the goal is to establish an assertion, the NEGATION of that assertion is taken as H_0, and the assertion itself is taken as H_1.

H_0 represents the default position: no difference exists between the groups or quantities compared. H_1 proposes that a difference does exist. For the ATM problem, H_0: $\mu = 270$ against H_1: $\mu < 270$. The original question — is there strong evidence for the claim? — is thereby translated into a question that can be answered: is there strong evidence for rejecting H_0?

H_0 always carries an equality, because only a definite number makes a distribution computable.

The Decision Rule

Once H_0 and H_1 are formulated, the sample data must select one of exactly two actions

ACTION 1

Reject H_0

The evidence against the default position is strong enough. The conclusion is that H_1 is substantiated by the data.

- A positive finding, supported by evidence

ACTION 2

Retain H_0

The evidence is not strong enough. The conclusion is that H_1 fails to be substantiated — which is not the same as concluding that H_0 is true.

- An absence of evidence, not evidence of absence

TESTS FOR μ WHEN n IS LARGE

Test Statistic, Rejection Region, Critical Value

01

Test statistic

The random variable whose value determines the action. For the ATM problem it is the sample mean $\bar{x}$.

02

Rejection region

The set of outcomes for which H_o is rejected. Here $R : \bar{x} \leq c$, where R stands for rejection of H_o.

03

Critical value

The boundary c of the rejection region. It is fixed before the data are seen, and it is what α determines.

A test is a rule stated in advance, not a judgement formed after looking at the number.

TESTS FOR μ WHEN n IS LARGE

Locating the Critical Value

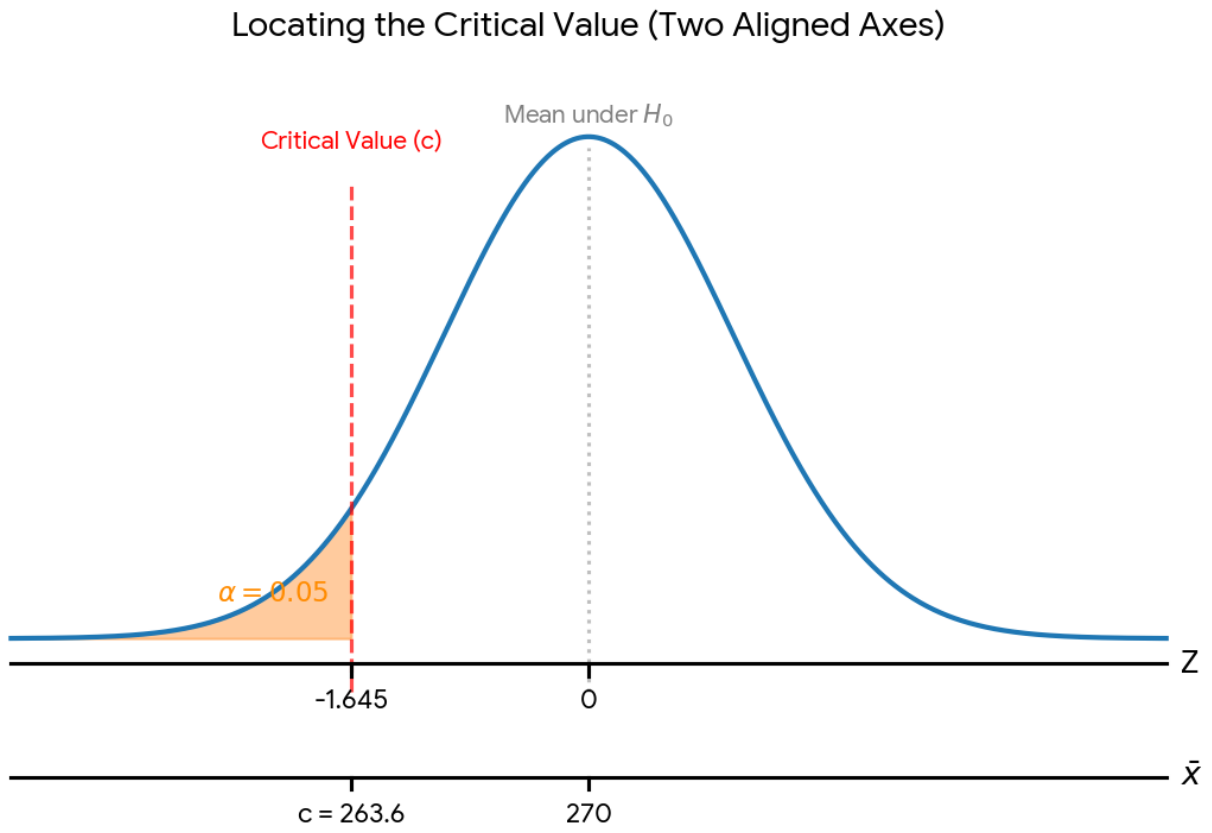
α fixes c, and c fixes the rejection region

THE ATM PROBLEM

Choose c so that P($\bar{x} \leq c$) = α when H_0 holds

Under H_0 the sample mean is approximately N(270, 24/√38). Standardising gives $Z = (\bar{x} - 270)/(24/\sqrt{38})$, and $P(Z \leq -1.645) = 0.05$.

- 24/√38 = 3.89 seconds — the standard error
- c = 270 − 1.645 × 3.89 = 263.60 seconds
- R : $\bar{x} \leq 263.60$, equivalently R : $Z \leq -1.645$



The Courtroom Analogy

H_0: the defendant is innocent. H_1: the defendant is guilty.

Verdict	Truth: innocent	Truth: guilty
Innocent	correct	Type II error
Guilty	Type I error	correct
The court never rules "innocent" — it rules "not proven"		

The Two Errors Named

Error	You have …	The result is …
Type I	rejected H_o when H_o is true	an innocent person convicted
Type II	failed to reject H_o when H_o is false	a guilty person released

Type I is the error of acting on something that is not there. Type II is the error of missing something that is.

A Worked Case: Testing a New Drug

H_0: the drug has no effect, and behaves like a placebo. H_1: the drug has an effect.

IF H_0 IS REJECTED

The conclusion is that the drug works. If H_0 was in fact true, this is a Type I error: a useless drug is adopted, and patients are treated with it.

IF H_0 IS RETAINED

The conclusion is that the drug does not work. If H_1 was in fact true, this is a Type II error: a genuinely effective drug is abandoned.

A Mnemonic: The Smoke Detector

TYPE I

False positive — a false alarm

The detector sounds when there is no fire. Something was declared present when it was absent.

- Báo động giả

TYPE II

False negative — a missed event

The detector stays silent while the house burns. Something real was present and went undeclared.

- Bỏ sót thật

α and β

α = P(reject H_0 | H_0 is true) — the level of significance. β = P(fail to reject H_0 | H_1 is true).

For the ATM problem these read as $\alpha = P(\bar{x} \le c \text{ when } \mu = 270)$ and $\beta = P(\bar{x} > c \text{ when } \mu < 270)$. False rejection of H_0 is ordinarily treated as the more serious of the two, which is why the threshold is placed where it is: a decision to reject H_0 must rest on strong evidence. α is chosen by the analyst; β is not chosen, and follows from α, from n, and from the true size of the effect.

Consequences of Each Error

TYPE I — FALSE POSITIVE

Something is believed significant when it is not

A medical test diagnoses cancer in a healthy patient, leading to treatment that is unnecessary and itself harmful. An investor acts on a false signal and takes a loss.

- Controlled directly, by the choice of α

TYPE II — FALSE NEGATIVE

Something real is missed

A security system fails to detect an intruder. A capable student is graded low through a marking error, and their progress is held back.

- Not controlled directly: it depends on n and on the effect size

Lowering α reduces false positives and increases false negatives. The two move in opposite directions.

The Trade-Off Between the Two Errors

No choice of the cutoff c minimises both α and β. Moving c to make α smaller makes β larger, and the reverse.

The cutoff is a single number on a single axis, and the two error probabilities are measured on opposite sides of it. Practice therefore fixes α at a conventionally small value — 0.10, 0.05 or 0.01 — and accepts whatever β follows. The only way to reduce both at once is to increase n, which narrows both distributions and is the reason sample size is worth paying for.

Example: Testing the ATM Claim

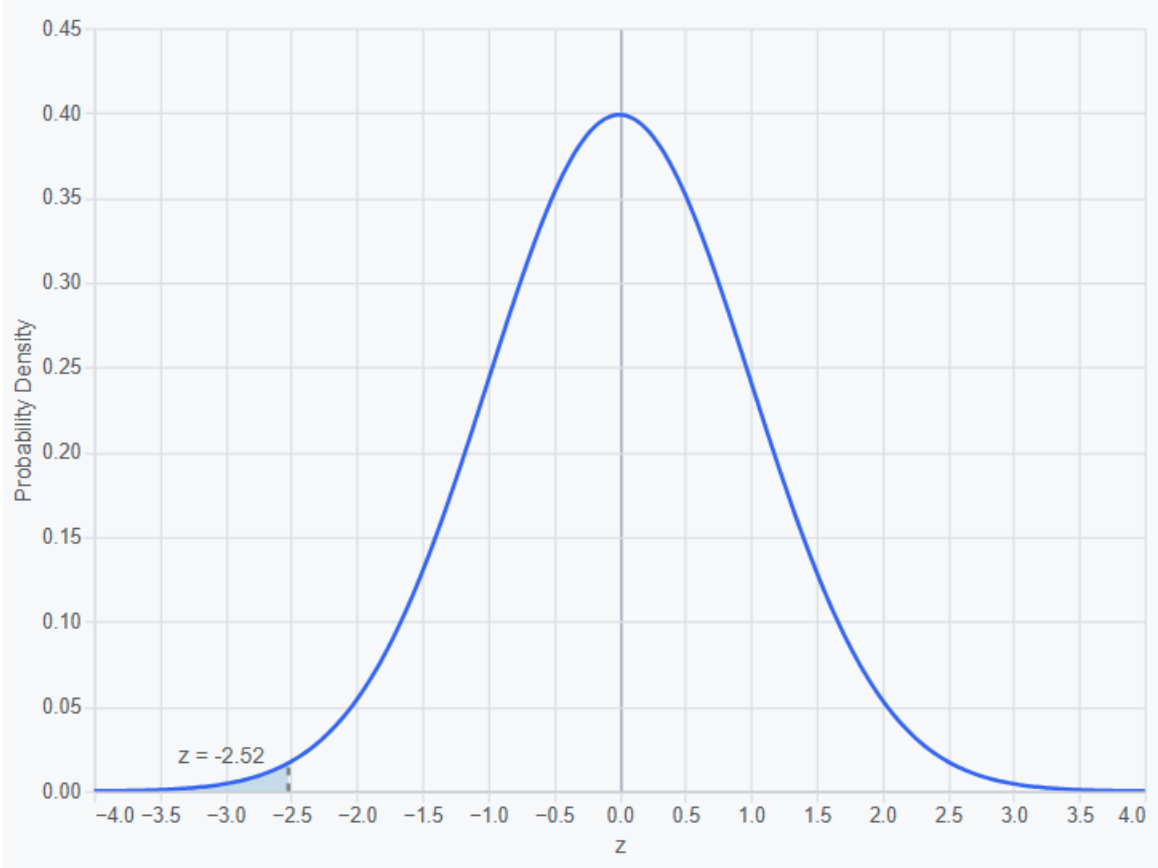
n = 38 · sample mean 261 seconds · sample standard deviation 22 seconds

H_0: M = 270 VS H_1: M < 270

At α = 0.025 the rejection region is R : Z ≤ −1.96

Z = (x̄ − 270)/(s/√38) = (261 − 270)/(22/√38) = −9/3.57 = −2.52, which lies inside R.

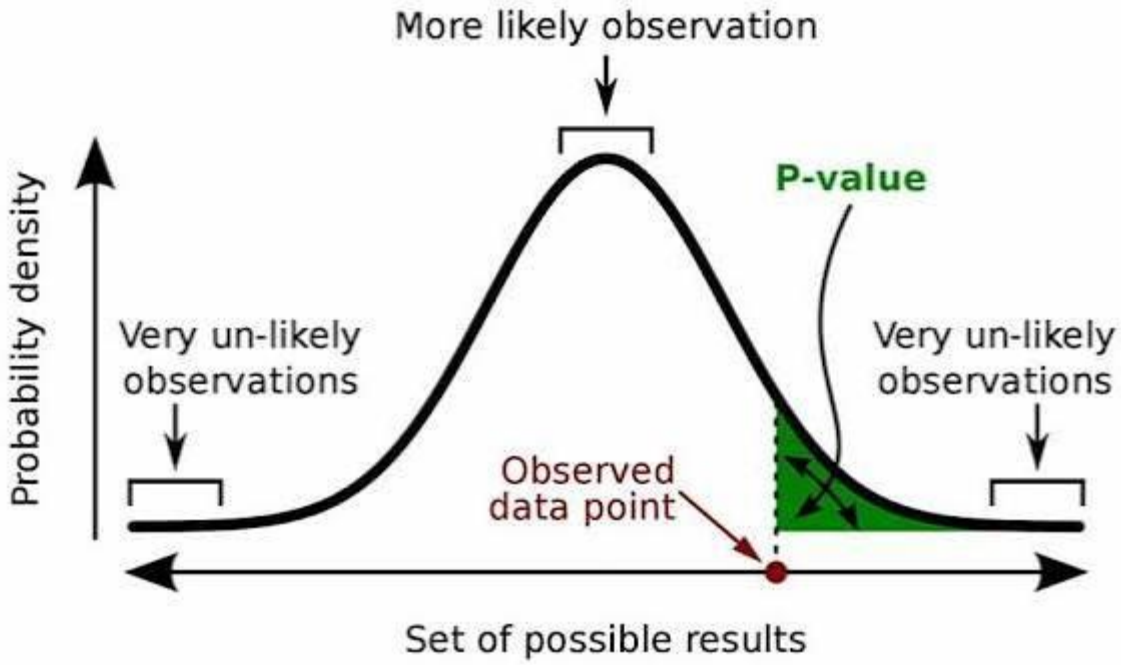
- −2.52 < −1.96, so H_0 is rejected
- The claim of a reduction in mean transaction time is strongly supported



The p-value

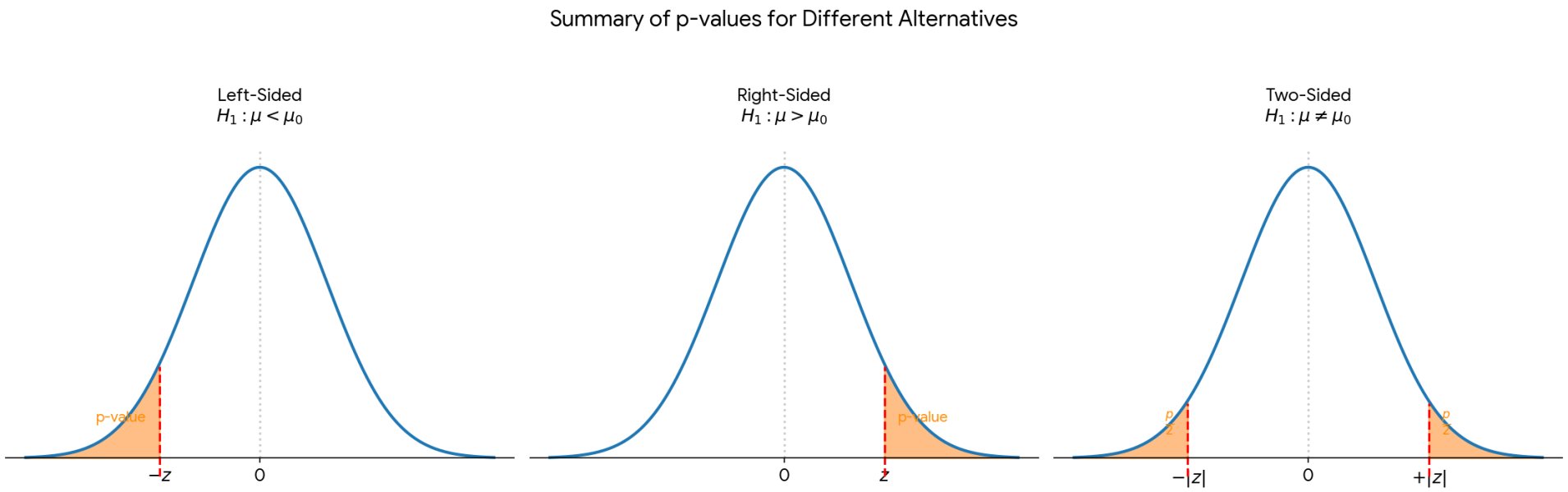
The p-value is the probability of obtaining a test statistic AS EXTREME AS OR MORE EXTREME THAN the one observed, GIVEN THAT H_0 IS TRUE.

It measures how well the sample evidence agrees with the null hypothesis. A small p-value means the observed result would be unlikely if H_0 held, and the evidence for rejecting H_0 is therefore strong. It does not report the probability that H_0 is true or false: it reports only how likely a result like this one would be in a world where H_0 holds.



p is a statement about the data under an assumption, never a statement about the assumption itself.

The p-value


Summary of p-values for Different Alternatives
Left-Sided
$H_1 : \mu < \mu_0$
p-value
$-z$
0
Right-Sided
$H_1 : \mu > \mu_0$
p-value
0
z
Two-Sided
$H_1 : \mu \neq \mu_0$
$\frac{p}{2}$
$\frac{p}{2}$
$-|z|$
0
$+|z|$

What a p-value Reports

01

It measures agreement with H_0

It states how compatible the sample evidence is with the null hypothesis, on a scale from 0 to 1.

02

Small p means strong evidence

The smaller the p-value, the less likely a result like this would be if H_0 held, and the stronger the case for rejecting H_0.

03

It is not P(H_0 is true)

The probability is computed inside the assumed world. It says nothing about how likely that world is.

04

It is not the size of the effect

A very small p-value can accompany a difference too small to matter. Sample size moves p; it does not move the difference.

Deciding with a p-value

The p-value is computed from the distribution of the test statistic, then compared with α

Comparison	Decision	What it means
$p \le \alpha$	reject H_0	a result like this is unlikely if H_0 holds; the evidence against H_0 is sufficient
$p > \alpha$	do not reject H_0	the evidence is insufficient — which is not a demonstration that H_0 is true

α is fixed before the data are examined. Comparing p with an α chosen afterwards is not a test.

A Worked p-value: A Headache Drug

H_0: the new drug relieves pain no better than the old one. H_1: the new drug is better.

The trial returns p = 0.03. At α = 0.05, since 0.03 < 0.05, H_0 is rejected: there is evidence that the new drug relieves pain better.

Read the number carefully. p = 0.03 states that IF the two drugs were equally effective, a difference at least as large as the one observed would appear in about 3 trials out of 100. Because that is unlikely, the assumption of equal effectiveness is doubted. The figure 0.03 is not the probability that the drugs are equally effective.

TESTS FOR μ WHEN n IS LARGE

How Strong Is the Rejection in the ATM Test?

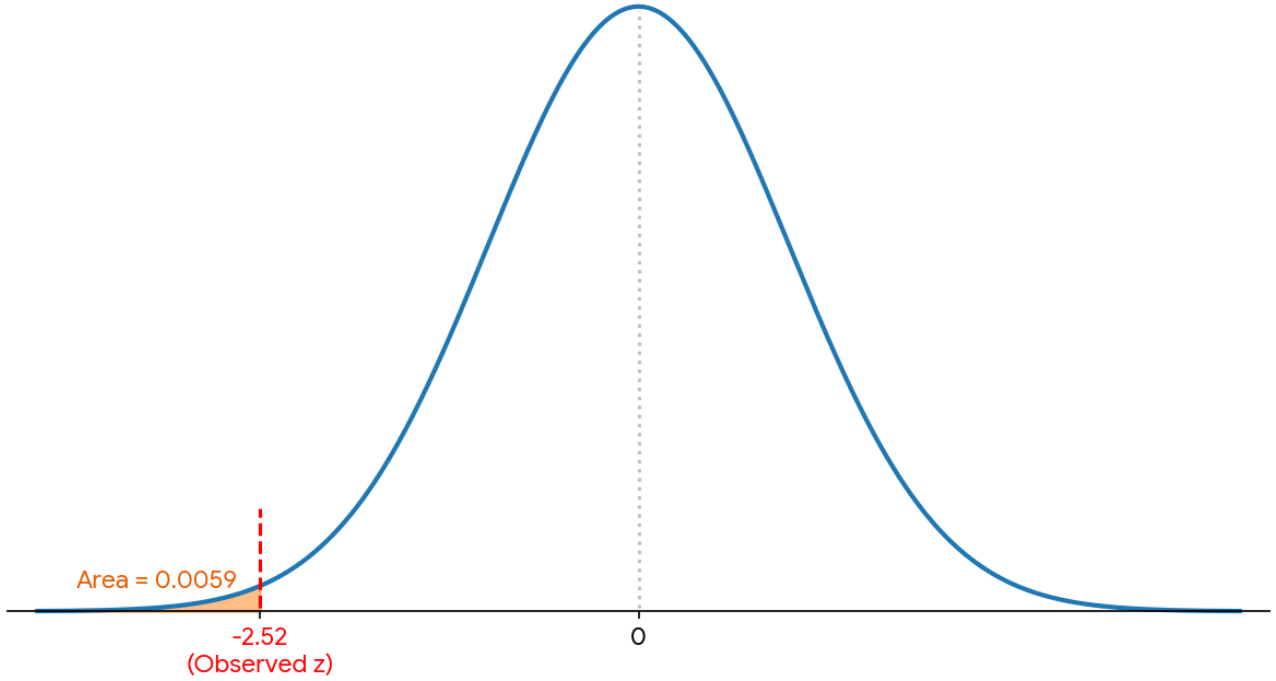
The smallest α at which H_0 would still be rejected

FROM Z = −2.52

$P(Z \leq -2.52) = 0.0059$

This significance probability is the p-value of the observed z. Compared with α = 0.025 it is far smaller, so H_0 is not merely rejected: it is rejected decisively, and the result is highly significant.

- H_1: $\mu < \mu_0$ is a left-sided alternative — R : $Z \leq -z(\alpha)$
- H_1: $\mu > \mu_0$ is a right-sided alternative — R : $Z \geq z(\alpha)$
- H_1: $\mu \neq \mu_0$ is two-sided — both tails, each of size α/2



The Five Steps for Testing a Hypothesis

STEP 1

Formulate the hypotheses

State H_0 and H_1. The assertion to be established goes in H_1; its negation goes in H_0.

STEP 2

State the test criterion

Name the test statistic and the form of the rejection region — left, right, or two-sided.

STEP 3

Determine the rejection region

With α specified, look up the critical value and write the region out.

STEP 4

Compute the test statistic

Substitute the sample figures into the statistic named in step 2.

STEP 5

Draw and interpret the conclusion

State whether H_0 is rejected at the specified α, in the language of the problem.

Steps 1 to 3 are completed before the data are looked at. Only steps 4 and 5 use the sample.

Example: A Weight-Loss Programme

A diet programme states that participants lose over 22 pounds in 5 weeks. From 56 participants, the sample mean is 23.5 pounds and the sample standard deviation is 10.2 pounds.

Is the statement substantiated by these findings? Test at a level of significance of 0.05, then compute the p-value and interpret the result. The figures to carry through the five steps: $n = 56$, $\mu_0 = 22$, $\bar{x} = 23.5$, $s = 10.2$, $\alpha = 0.05$.

The sample mean is 23.5, above the claimed 22. Is that enough to substantiate the claim?

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TESTS FOR μ WHEN n IS LARGE

The Weight-Loss Test, Step by Step

STEP 1

Hypotheses

$H_0 : \mu = 22$ versus $H_1 : \mu > 22$

STEP 2

Test statistic

$Z = (\bar{x} - 22) / (s/\sqrt{n})$ — a right-sided alternative, so R : $Z \geq c$

STEP 3

Rejection region

$z(0.05) = 1.645$, so R : $Z \geq 1.645$

STEP 4

Compute

$Z = (23.5 - 22) / (10.2/\sqrt{56}) = 1.5 / 1.36 = 1.10$, which is not in R

STEP 5

Conclude

H_0 is not rejected at $\alpha = 0.05$. The claim that $\mu > 22$ is not substantiated.

$p = P(Z \geq 1.10) = 0.1357$ — the smallest α at which H_0 could be rejected. It is not negligible, so the data give no strong basis for rejection.

TESTS FOR μ WHEN n IS SMALL

Large Samples and Small Samples

LARGE n

The central limit theorem does the work

For a large sample the distribution of $\bar{x}$ is approximately normal whatever the population looks like. Nothing has to be assumed about the population itself.

- Test statistic: Z, read from the standard normal table

SMALL n

The theorem no longer applies

Below about n = 30 the approximation fails, and the appropriate method depends on what can be assumed about the population distribution.

- If the population is normal: test statistic T, with n − 1 degrees of freedom

The five steps do not change. Only the statistic and the table change.

Example: Bacteria Count at a Lake Beach

A city health department wishes to determine whether the mean bacteria count per unit volume of water is within the safety level of 200. Ten unit-volume samples are collected.

The counts are 175, 190, 215, 198, 184, 207, 210, 193, 196 and 180. Do the data strongly indicate that there is no cause for concern? Ten observations is far below the threshold at which the central limit theorem can be relied on, so this is a small-sample problem and the t distribution applies.

TESTS FOR μ WHEN n IS SMALL

The Bacteria Count Test

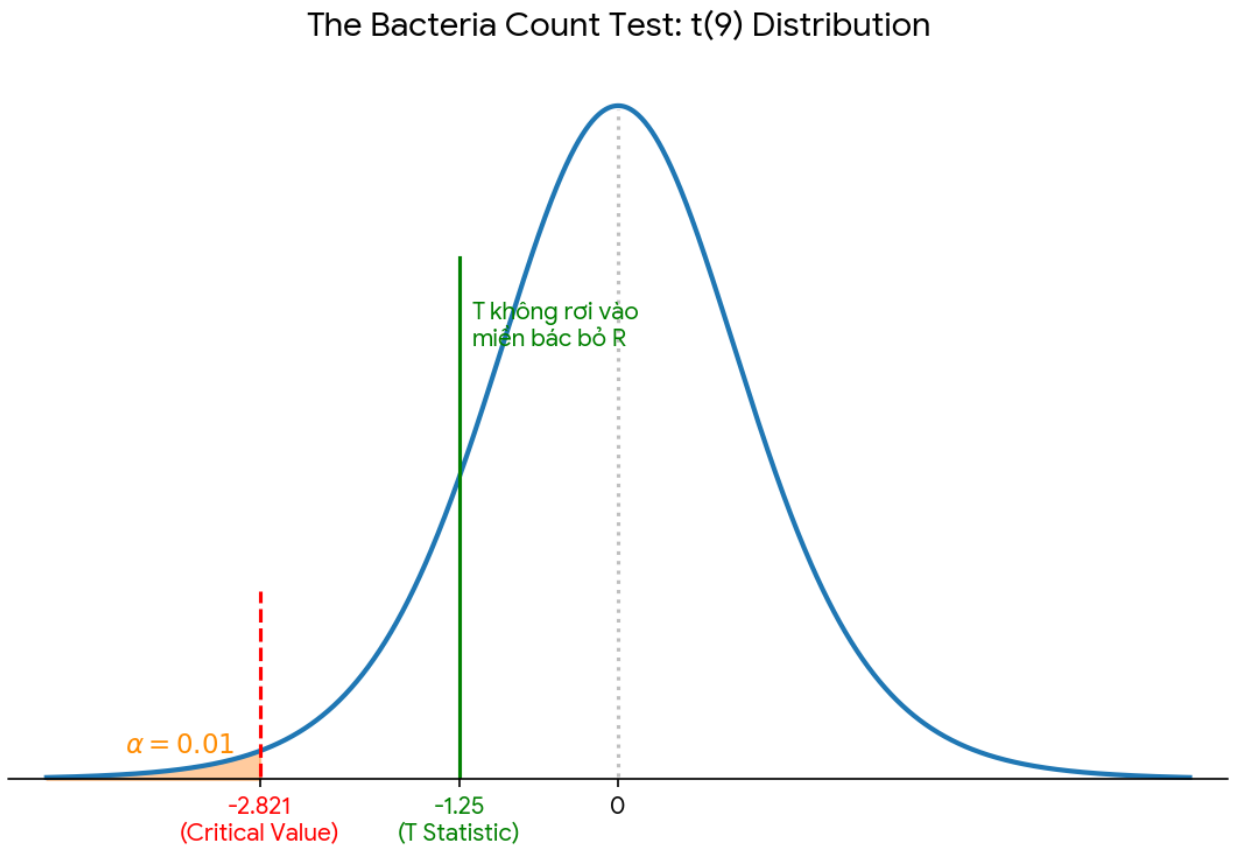
$H_0 : \mu = 200$ versus $H_1 : \mu < 200$ · $\alpha = 0.01$ · left-sided

$T = (\bar{X} - 200) / (S/\sqrt{10})$

With df = 9, the rejection region is R : $T \leq -2.821$

From the ten measurements, $\bar{x} = 194.8$ and $s = 13.14$, so $T = (194.8 - 200)/(13.14/\sqrt{10}) = -5.2/4.156 = -1.25$. This does not fall in R.

- H_0 is not rejected at $\alpha = 0.01$
- There is no strong evidence that the true mean is within the safety level



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Testing a Population Proportion

$H_0 : p = p_0$ versus $H_1 : p \neq p_0$, with n large

$Z = (\hat{p} - p_0) / \sqrt{(p_0 q_0 / n)}$ where $\hat{p} = X/n$ is the sample proportion and $q_0 = 1 - p_0$.

For a large sample the sample proportion is approximately normally distributed. Under the null hypothesis the proportion has the specified value p_0, so the standardised statistic above has the N(0,1) distribution. Because the alternative is two-sided, the rejection region of a level-α test is $R : |Z| \geq z(\alpha/2)$. The five steps are unchanged; only the statistic differs.

Example: Families Below the Poverty Level

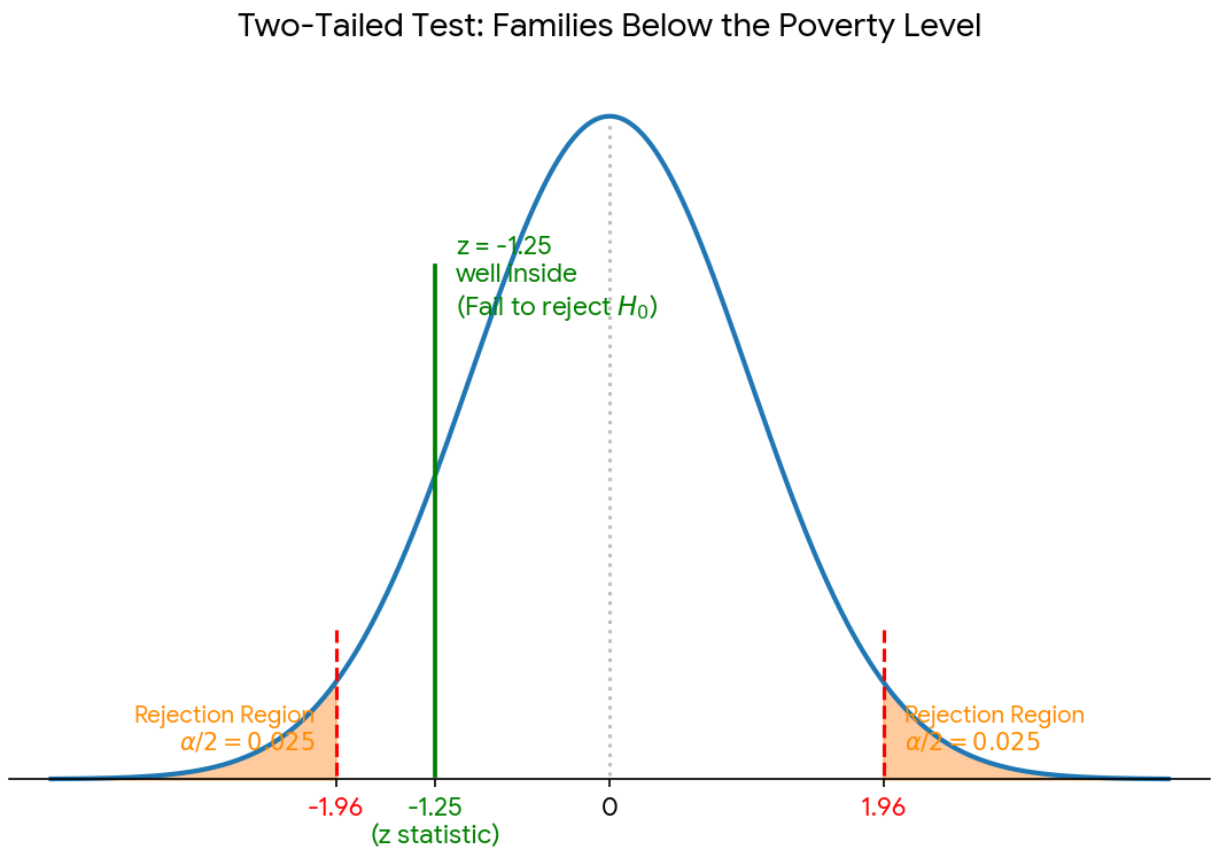
A census five years ago recorded 20%. A sample of 400 families now finds 70 below the level.

$H_0 : P = 0.20$ VS $H_1 : P \neq 0.20$

At $\alpha = 0.05$ the rejection region is R : $|Z| \geq 1.96$

$\hat{p} = 70/400 = 0.175$. The standard error is $\sqrt{(0.2 \times 0.8 / 400)} = 0.020$, so $Z = (0.175 - 0.200)/0.020 = -1.25$.

- $|-1.25| = 1.25 < 1.96$, so H_0 is not rejected at $\alpha = 0.05$
- The data do not indicate that the percentage has changed



Three Things to Take Away

1. ONE PROCEDURE

Five steps, unchanged across means, proportions and variances. Only the test statistic and the table change.

2. FIXED IN ADVANCE

Hypotheses, the form of the region and α are settled before the data are seen. Only computation comes afterwards.

3. TWO WAYS TO BE WRONG

α is chosen and β follows. Tightening one loosens the other, and only a larger n reduces both.